

Customer Segmentation & Churn Pattern Analytics in European Banking

Research Paper

Abstract

Customer churn represents a significant challenge for retail banking because the loss of existing customers can reduce customer lifetime value, increase acquisition costs, and create revenue instability. This study presents a segmentation-driven analysis of customer churn in European banking using demographic, geographic, engagement, tenure, and financial characteristics. The analysis uses 10,000 customer records and examines churn across geography, age, gender, activity status, credit-score bands, tenure, balance, salary, and product usage. The overall observed churn rate is 20.37%. Germany records the highest observed churn rate at 32.44%, while customers aged 46–60 show particularly high churn, reaching 67.33% among German customers. Inactive customers show a 26.85% observed churn rate compared with 14.27% among active customers. High-balance customers have a 24.16% observed churn rate, and churned high-balance customers represent approximately €110.8 million in balance exposure. The findings demonstrate the value of segmentation analytics for identifying priority customer groups and supporting targeted retention strategies.

1. Introduction

Customer retention is an important concern in retail banking. Existing customers contribute to long-term relationships and financial value, while customer acquisition requires additional marketing and operational resources. Understanding customer churn can therefore support more focused retention strategies.

Traditional churn reporting often focuses on a single overall percentage. While useful, this does not explain which customer groups contribute most to churn or where financial exposure is concentrated. A segmentation-based approach provides a more granular view by comparing churn across geography, age, engagement, tenure, credit profile, balance, salary, and product usage.

This project develops a systematic customer segmentation and churn analytics framework for European banking customers. The analysis is implemented through exploratory data analysis and an interactive Streamlit dashboard.

2. Problem Statement

Despite having customer-level information, banks may face difficulty identifying high-risk customer segments, understanding how churn differs across regions and demographic groups, and quantifying the financial profile associated with churn. Without structured segmentation analytics, retention strategies may remain generic and reactive. This project addresses these challenges by analyzing churn patterns across multiple customer dimensions.

3. Objectives

- Measure the overall customer churn rate.
- Identify churn distribution across customer segments.
- Compare churn behaviour across France, Germany, and Spain.

- Understand churn among high-value and high-balance customers.
- Evaluate engagement and tenure patterns associated with churn.
- Compare demographic and financial profiles of retained and churned customers.
- Quantify balance exposure associated with churned high-value customers.
- Provide actionable insights and recommendations for targeted retention planning.

4. Dataset Description

The dataset contains 10,000 customer records and 14 original fields covering customer identity, geography, demographics, creditworthiness, tenure, account balance, product ownership, engagement, salary, and churn status.

The original analytical fields include CustomerId, Surname, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited. Exited is the churn indicator, where 0 represents a retained customer and 1 represents a churned customer. Surname was treated as non-analytical identity information.

5. Analytical Methodology

Data ingestion and validation were performed to confirm dataset structure, binary-variable consistency, categorical values, and missing values. The dataset contains 10,000 observations and no missing values were identified.

Data preparation included removal of non-analytical identity information and creation of derived segmentation fields. Age was grouped into <30, 30–45, 46–60, and 60+. Credit scores were grouped into Low, Medium, and High. Tenure was grouped into New, Mid-term, and Long-term customers. Balance was segmented into Zero-balance, Low-balance, and High-balance customers.

Churn rate was calculated as the proportion of customers with Exited = 1 within the relevant population or segment. This enables direct comparison of churn behaviour across customer groups.

6. Exploratory Data Analysis and Findings

The dataset contains 7,963 retained customers and 2,037 churned customers, producing an overall observed churn rate of 20.37%.

Geographic analysis shows that Germany has the highest observed churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain. Germany has a Geographic Risk Index of 1.59, while Spain and France have indices of 0.82 and 0.79 respectively.

The geography-age interaction shows that customers aged 46–60 have the highest observed churn rate in all three countries: 45.79% in France, 67.33% in Germany, and 40.66% in Spain. The German 46–60 segment is therefore the strongest observed geography-age churn concentration.

Female customers have an observed churn rate of 25.07%, compared with 16.46% for male customers. This is an observed association and should not be interpreted as evidence that gender causes churn.

Inactive customers show a 26.85% observed churn rate compared with 14.27% among active customers, indicating a strong association between lower engagement and higher observed churn.

Financial comparison shows that churned customers have a higher average balance (€91,108.54) than retained customers (€72,745.30). Churned customers also have a slightly higher average estimated salary (€101,465.68 versus €99,738.39) and slightly fewer products (1.48 versus 1.54). Average credit scores are relatively close (645.35 versus 651.85).

High-balance customers have a 24.16% observed churn rate, while low-balance customers have a similar 24.00% rate and zero-balance customers have a lower 13.82% rate. Thus, balance level alone should not be treated as a causal explanation.

Salary-group churn rates range from 19.80% to 21.56%, showing only moderate variation across salary groups.

Germany contributes 39.96% of all churned customers, France contributes 39.76%, and Spain contributes 20.27%. This demonstrates why both churn rate and segment size should be considered.

Churned high-balance customers represent approximately €110.8 million in balance exposure. This is balance exposure rather than direct revenue loss because account balance is not equivalent to bank revenue.

7. Key Performance Indicators

Overall Churn Rate: 20.37%

High-Value Churn Ratio: 24.16%

Germany Geographic Risk Index: 1.59

Inactive Customer Churn Rate: 26.85%

High-Balance Churn Exposure: approximately €110.8 million

8. Business Insights and Recommendations

1. Prioritize Germany: Germany has the highest observed churn rate and geographic risk index, so retention analysis should receive particular attention in this geography.
2. Focus on customers aged 46–60: This group has the highest observed churn across all three countries, especially in Germany.
3. Improve engagement: Inactive customers show substantially higher observed churn, suggesting that engagement-focused initiatives may be valuable.
4. Protect high-value relationships: High-balance churners represent approximately €110.8 million in balance exposure and should be monitored closely.
5. Use targeted retention strategies: Geography, age, engagement, and financial profile should be considered together rather than relying on generic retention campaigns.
6. Monitor both rate and contribution: Germany and France each contribute approximately 40% of total churn, demonstrating why both segment size and churn rate matter.

9. Streamlit Dashboard

An interactive Streamlit dashboard was developed to operationalize the analysis. It provides Churn Overview, Customer Segmentation, Churn Pattern Analysis, and Business Insights & Recommendations sections. Users

can filter by geography, gender, activity status, and age group. Dynamic KPIs and customer drill-down views update according to the selected filters.

10. Conclusion

This project demonstrates how segmentation-driven exploratory analytics can provide a detailed understanding of customer churn in European banking. The overall observed churn rate is 20.37%, but churn is not distributed uniformly across customer groups. Germany, inactive customers, the 46–60 age group, and high-balance customers represent important areas for closer attention. The geography-age analysis shows that German customers aged 46–60 have the highest observed churn rate at 67.33%. Churned high-balance customers represent approximately €110.8 million in balance exposure. The resulting Streamlit dashboard converts these findings into an interactive analytical tool for targeted, data-informed retention planning.

11. Limitations

The analysis identifies associations and observed patterns rather than causal relationships. Balance exposure should not be interpreted as direct revenue loss. The project focuses on descriptive and segmentation-driven analytics and does not establish causal mechanisms for customer churn.

12. References

European Central Bank. Banking and financial stability information. <https://www.ecb.europa.eu/>
Unified Mentor. Project and dataset context provided for Customer Segmentation & Churn Pattern Analytics in European Banking.
Customer churn dataset used in the project and processed through the project's data preparation workflow.

KPI Summary Table

KPI	Result	Meaning
Overall Churn Rate	20.37%	Observed churn across all customers
High-Value Churn Ratio	24.16%	Observed churn among high-balance customers
Germany Geographic Risk Index	1.59	Germany churn relative to overall churn
Inactive Customer Churn Rate	26.85%	Observed churn among inactive customers
High-Balance Churn Exposure	€110.8M	Balance associated with churned high-balance customers